

Shri G.S. Institute of Technology and Science, Indore M.P.

EE 10510 Basic Electronics Engineering

PRACTISE SHEET- 2

1. Do the following conversion:

(i) $(1010111)_2 = (\text{-----})_{10}$ (ii) $(537)_8 = (\text{-----})_{16}$ (iii) $(521)_{10} = (\text{-----})_8$

(iv) $(1101101)_2 = (\text{-----})_{\text{gray}}$ (v) $(4CA5)_{16} = (\text{-----})_{10}$

2. Using Boolean algebra techniques, simplify the following expressions and realize the minimize expression using basic logic gates.

(i) $AB + A(B+C) + B(B+C)$

(ii) $ABCD + AB(CD)' + (AB)'CD$

3. What are universal logic gates? Implement AND and OR gates using universal gates.

4. Simplify the following expressions using K-map and implement circuit diagram using gates.

(i) $A'B'C' + A'B'C + AB'C + ABC + ABC'$

(ii) $F(A,B,C) = \sum m(0,2,3,4,5,6)$

5. Prove that

i. If $A \oplus B = 0$ then $A = B$

ii. $A \oplus B = A' \oplus B'$

iii. $(A \oplus B)' = A' \oplus B = A \oplus B'$

6. What are combinational circuits? Explain working of half adder.

7. Describe the operation of 3 bit adder with truth table, expression and circuit diagram.

8. Implement full adder using two half adder.